

Thermal expansion

$$\Delta l = \alpha l_0 \Delta T$$

$$\Delta A = 2\alpha A_0 \Delta T$$

$$\Delta V = \beta V_0 \Delta T = 3\alpha V_0 \Delta T$$

Calorimetry

$$Q = mc\Delta T = nC\Delta T \text{ (no phase transition)}$$

$$Q = mL \text{ (for a phase transition)}$$

$$\sum_{i=1}^n Q_i = 0 \text{ (for an isolated system)}$$

Heat conduction

$$H = \frac{dQ}{dt} = -kA \frac{dT}{dx}$$

Ideal gases

$$PV = Nk_B T = nRT$$

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

$$v_{rms} = \sqrt{\overline{v^2}} = \sqrt{\frac{3k_B T}{m}}$$

$$\Delta E_{int} = Q - W$$

$$dE_{int} = dQ - PdV$$

$$W = \int PdV \text{ (for reversible processes)}$$

$$E_{int} = \frac{d}{2} Nk_B T = nC_V T$$

$$C_V = \frac{d}{2} R$$

$$C_P - C_V = R = N_A k_B$$

$$\gamma = \frac{C_P}{C_V} = \frac{d+2}{d}$$

$$e = \frac{W_{net}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

$$e_{ideal} = 1 - \frac{T_L}{T_H}$$

$$\Delta S = \int \frac{dQ}{T} \text{ (for reversible processes)}$$

$$dQ = TdS$$

$$\Delta S_{syst} + \Delta S_{env} > 0 \text{ (for irreversible processes)}$$

$$\Delta S_{syst} + \Delta S_{env} = 0 \text{ (for reversible processes)}$$

Integrals

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n! 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^\infty x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int (1+x^2)^{-\frac{1}{2}} dx = \arctan(x)$$

$$\int (1+x^2)^{-\frac{3}{2}} dx = \frac{x}{\sqrt{1+x^2}}$$

$$\int \frac{1}{x} dx = \ln(x)$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2)$$

$$\int \frac{1}{\cos(x)} dx = \ln \left(\left| \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right| \right)$$

$$\int \frac{1}{\sin(x)} dx = \ln \left(\left| \tan \left(\frac{x}{2} \right) \right| \right)$$

Taylor Expansions near x=0

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha-1)\alpha}{2} x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Trigonometry

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 2 \cos^2(x) - 1$$

$$\sin(a+b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$\cos(a+b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$1 + \cot^2(x) = \csc^2(x)$$

$$1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta) \text{ (law of cosine)}$$

	Q	W
Isobaric ($P=const$)	$nC_p\Delta T$	$P\Delta V$
Isochoric ($V=const$)	$nC_v\Delta T$	0
Isothermal ($PV=const$)	$nRT \ln\left(\frac{V_f}{V_0}\right)$	$nRT \ln\left(\frac{V_f}{V_0}\right)$
Adiabatic ($PV^\gamma = const$)	0	$-\frac{d}{2}(P_f V_f - P_0 V_0)$

C_v, C_p and $\gamma = C_p/C_v$ for an Ideal Gas

1. For a monoatomic gas, $d=3$

$$C_v = \frac{3R}{2}, C_p = \frac{5R}{2} \text{ and } \gamma = \frac{5}{3} = 1.67$$

2. For a diatomic gas, $d=5$

$$C_v = \frac{5R}{2}, C_p = \frac{7R}{2} \text{ and } \gamma = \frac{7}{5} = 1.4$$

3. For a triatomic or polyatomic gas, $d=6$

$$C_v = 3R, C_p = 4R \text{ and } \gamma = \frac{4}{3} = 1.33$$

Areas and volumes

$$\text{Cylinder: } A = 2\pi RH ; V = \pi R^2 H$$

$$\text{Cylindrical shell: } dV = 2\pi R H dR$$

$$\text{Sphere: } A = 4\pi R^2 ; V = \frac{4}{3}\pi R^3$$

$$\text{Spherical shell: } dV = 4\pi R^2 dR$$